



Understanding an “In Principle Rather Old Gas Discharge”: Plasma Diagnostics on Dielectric Barrier Discharges

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Abstract

Dielectric barrier discharges are one of the most applied and investigated atmospheric pressure plasma source, but also a challenge for plasma diagnostics. Here, a brief historical overview of diagnostics of the physical behavior and basis plasma parameters for barrier discharges is provided by mentioning selected highlights over the last about ninety years and the discussion of some developments after the first application of the diagnostic methods. Furthermore, current issues in research are addressed. Rather than focusing on the methodology of the respective diagnostics, it aims to summarize the progress made in the field on barrier discharges in particular, and to highlight the challenges posed by this type of gas discharge.

Keywords Nonthermal plasma · Atmospheric pressure plasma · Dielectric barrier discharge · Plasma technology · Plasma diagnostics

Introduction

The title of this article is motivated by two considerations. Firstly, dielectric barrier discharges or dielectric barrier discharges (DBDs), for a long time also named silent discharges, have been known for over 150 years [1, 2]. Secondly, the title is borrowed from a quote in an article by Kogelschatz [3]. In this well-received review article from 2003, the author summarizes and argues: “Industrial applications of dielectric-barrier discharges have come a long way and, more recently, have substantially increased their market penetration. No doubt, this trend will continue. Several reasons can be given for this renaissance of an **in principle rather old gas discharge**.”

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This statement has proven to be true over the last 20 years, and the “renaissance” he refers to is still ongoing. As summarized in a recent update-paper of this article, in addition to its traditional applications in ozone synthesis and surface activation, this type of discharge has also found use in biomedical applications, light sources, exhaust air cleaning, analytical devices, and surface coating as summarized recently in [4]. Current research focuses on gas treatment including carbon dioxide conversion, plasma chemical synthesis and catalysis, combustion initiation [5], flow control, as well as liquid treatments [6–12]. In the relatively new field of plasma agriculture, it will certainly also find many applications as a simple, robust, nonthermal atmospheric-pressure plasma source [13–15]. Furthermore, interest in partial discharges in insulated high-voltage systems (i.e. “undesired barrier discharges” [16]), is ongoing, particularly given the trend towards smaller insulated systems and electrical circuits.

This article aims to provide a historical overview of diagnostics on the discharge development and plasma parameters for this type of discharge and to highlight some of the current issues in research. Rather than focusing on the methodology of the respective diagnostics, it aims to summarize the progress made in the field on barrier discharges in particular, and to highlight the challenges posed by this type of plasma source. As the focus is on the discharge physics and the determination of basic plasma parameters such as the electric field strength and the number densities of active species, most of the selected highlights were obtained on relatively simple discharge geometries, such as plane-parallel electrode arrangements or pin-to-pin configurations (see Fig. 2). In particular the single- or mono-filament arrangements enabled the distinct study of the plasma in its most elemental form. A broad overview on the possible other electrode configurations for dedicated applications of DBDs was already given earlier [17]. In the last two decades more complex geometries like coplanar DBDs, plasma jets or microcavity arrays were studied.

The measurement methods themselves are presented in other review articles and not in the scope of this contribution, which emphasizes on the historical development about the study of the discharge mechanisms and the determination of basic plasma parameters. The reader is referred to this literature. For example, in reference [18] the diagnostics for microplasmas and microdischarges are summarized also including results beyond barrier discharges. Here, a short update about the efforts on barrier discharges within the last 12 years is given. The articles [19–24] describe optical measurement methods for atmospheric pressure plasmas in general or in interaction with surfaces, including laser diagnostics. Methods for recording discharge development are demonstrated and summarized in [17, 25], while reference [26] deals particularly with the measurement of the electric field strength as a basic plasma parameter, also beyond barrier discharges. Here, the Special Issue on “Best Practices for Atmospheric Pressure Plasma Diagnostics” [27] deserves a special mention. Furthermore, the immense progress that modelling has contributed over the last three to four decades should be emphasized here as well, see e.g. the contributions from Eliasson et al. [28, 29], Pietsch et al. [30–32], Kushner et al. [33–37], Boeuf et al. [38–40], Bogaerts et al. [41–44], Becker et al. [45–49] and Popov et al. [50–52]. The investigation of the plasma chemistry by dedicated gas analysis (e.g. FTIR, GC and GC-MS) is outside the scope of this article.

In line with the focus of this article, the next section only briefly discusses the principle and functioning of barrier discharge as a more detailed introduction including the discharge regimes and the underlying mechanisms is provided elsewhere, e.g. in [17, 53, 54].

However, the physical behavior of the barrier discharge gives rise to the aforementioned challenges. This is followed by the brief historical overview which will only cover a few selected highlights, such as the fast electrical measurements, determination of species densities, temperatures and electric field strength as well as surface charges on insulating barriers. This overview is by no means complete. It should only illustrate the difficulties, but also the progress in this field and how it has motivated further investigations in the last decade. The final section aims to identify potential areas for future research and development.

The Barrier Discharge Principle

The defining feature of barrier discharge is the presence of at least one insulator (usually dielectric, but ferroelectric is also possible) within the discharge gap, which plays a key role in generating distinctly nonthermal plasma at elevated pressures of around 1 atmosphere or higher. At sufficiently high voltage, a self-sustained gas discharge occurs due to multiple electron avalanche generations. As the generated charge carriers cannot flow through the insulator, they accumulate on its surface forming a surface charge which is creating a field that is opposite to the applied electric field (see Fig. 1). Consequently, the total electric field strength decreases and the discharge terminates automatically. This inherent limitation of discharge duration locally restricts the current, and thus the coupled plasma power density. Despite the fact that barrier discharges can now be found in many different geometries and applications (see, for example [17]), it is always this principle that prevents the transition into a spark discharge, i.e. causing so-called partial breakdown. This gives the plasma its distinctly nonthermal character, which does not require sophisticated power control by, for example, external circuit measures or gas flows.

Due to operation at elevated pressure the gas gap distances are kept small — usually in the millimetre range — in accordance with Paschen's law [55] to allow operation with moderate high-voltage amplitudes (up to around 30 kV). The increased pressure also favours streamer breakdown leading to a filamentary plasma. Only under specific conditions relating to gas mixtures, voltage forms and/or electrode materials (particularly with regard to the insulator, or the “barrier”), diffuse or uniform discharge regimes are obtained [56–59].

Filamentary discharge operation in particular has presented and continues to present specific challenges for experimenters. A filament can be regarded as a repeatedly generated transient microdischarge with duration for only a few tens of nanoseconds to microseconds, depending on the operation parameters, gas composition, pressure and the electrode

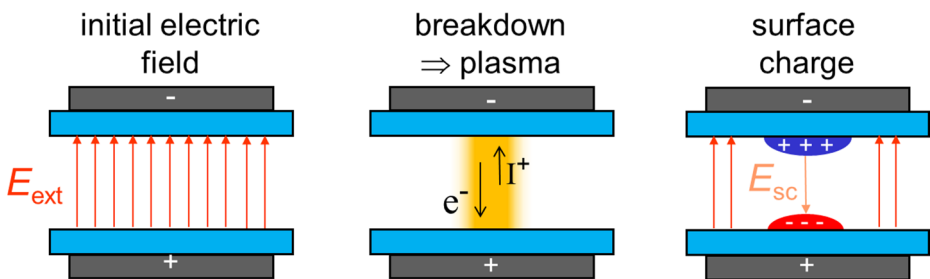


Fig. 1 Scheme about the main role of the barrier in DBDs (E_{ext} – electric field by applied voltage; e^- – electrons; I^+ – positive ions; E_{sc} – opposed electric field by surface charges)

geometry. Typically, many filaments are distributed erratically in space and time over the electrodes area. Figure 2 (a) shows an example. A total of ten filaments are clearly discernible within the 2 cm wide lateral discharge gas gap, while the current pulses measured in a single-filament arrangement (see Fig. 2 (b)) demonstrate a duration that is dependent on the gas, and ranging a few tens of nanoseconds. Another challenge lies in their size and morphology. Microdischarges have a diameter of approximately 100 μm , but on the insulator surface, they form discharge channels of a similar diameter, some of which are branched. In case of surface DBDs both electrodes are in contact with the barrier and the microdischarges can fully propagate along the gas-dielectric interface. The final challenge of diagnostics is the low current, charge (typically between 0.1 and 1 nC) and energy (typically 1–10 μJ), which may reflect in low reactive particle densities and weak radiation emission for diagnostic purposes.

Brief Historical Overview of Diagnostics at Barrier Discharges

As early as 1777, Lichtenberg produced a gas discharge in ambient air on a plate made of an electrically insulating material. He used dust particles to make the surface charges remaining after the discharge visible [61]. His famous “Lichtenberg dust figures” clearly demonstrate the influence of polarity, a phenomenon that was also confirmed by modern methods. The first ozone generator was invented by Siemens in 1857 [62], and this marked the beginning of DBDs as a widely used technological plasma sources. Just a few years after Siemens’ work, the first drinking water treatment plants with ozone generators came into operation [3, 63–65]. Already in 1934 Darrah patented the application of DBDs for the treatment of liquids and textiles via oxidizing processes [66]. However, since ozone generation was the most significant application for many decades, research on DBDs focused on air, oxygen, and N_2/O_2 mixtures. The fact that the first book dedicated on barrier discharges by Samoilovich, Gibalov and Kozlov focuses primarily on ozone synthesis is no coincidence [67] and the book by Becker, Schoenbach, Kogelschatz and Barker [54] even includes the term “air plasmas” in the title, although other gas mixtures (and types of plasma) are discussed too. However, the explanations of the discharge physics in these books are, of course, universally applicable. Some of the milestones presented in these references are presented

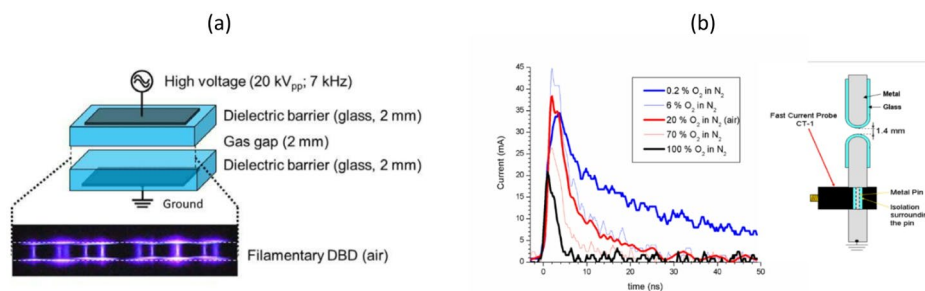


Fig. 2 (a) Example for a filamentary barrier discharge in air in a symmetric volume DBD arrangement (about 2 cm wide; left) and (b) current pulses measured for single microdischarges in different N_2/O_2 gas mixtures [60] with the corresponding electrode-current probe arrangement (right)

in Fig. 3. In the following discussion, these points will be addressed sequentially and later achievements will be mentioned as well while a concise synopsis of ensuing advancements will also be provided.

Despite its early application in ozone synthesis, but also for other chemistries, it was not until the mid-1930s that sufficiently fast oscilloscopes became available to determine the filamentary discharge character [68, 79, 80]. The Fig. 3 (a) shows the footprints of the microdischarge channels on a photographic plate by Buss [68] together with an current oscillogram presented by Klemenc et al. in 1937 [79]. In the 1940s, Manley published his work on electrical characterisation and power determination [69], which is still applied to determine the plasma power but has since been constantly expanded [81–87]. Figure 3 (b) shows the charge-voltage (QV) plot from Manley's original paper often called "Lissajous-Figures" in the DBD literature. In most cases it has the form of an parallelogram like in this example but meanwhile other forms has been investigated.

In the 1950s, DBDs opened up its second most important area of application with surface activation, and from the 1960s onwards, other gases besides N_2/O_2 mixtures were also investigated. These activities were driven by the emerging development of gas lasers, discharge lamps, and plasma displays [4]. Due to the very specific application areas, the research was conducted in different scientific communities. This phenomenon is particularly evident in the literature on such a widespread and versatile form of gas discharges as DBDs. It is important to acknowledge the efforts of Ulrich Kogelschatz in his review articles, where he demonstrated a commendable commitment to synthesizing these developments and enable a transfer of knowledge between the different communities.

The utilisation of oscilloscopes and image intensifier recordings in the study of barrier discharges can be traced back to the 1950s. Then, the 1980s can be regarded as the first "golden decade" in terms of DBD diagnostics and simulation. Notable achievements include the recordings of current pulses from individual microdischarges by Hirth and Kogelschatz [70, 88] (see Fig. 3 (c) for oxygen) and the determination of dissipated charge in microdischarges by Drimal et al. [72], see. Figure 3 (e) where a 2nd and 3rd maximum of the distribution histogram appears at integer multiples of the first peak, i.e. these peaks are the result of 2 or 3 simultaneous microdischarges.

The measurement of transient and fast current signals is not straightforward [89]. Even the application of Rogowski coils and current probes (or monitors) has its limitations. The authors in [88] stated "The leading edge is presumably not resolved by our measuring circuit which had a rise time of 1 ns." Although modern current monitors achieve sub-ns rise times (see below), the bandwidth of probes and oscilloscopes is constrained, and the external circuit inductance and parasitic capacity frequently cause supplementary oscillations, which disturb the measurement and the analysis. In the context of fast pulsed DBDs, it is important to consider a substantial displacement current. This is attributable to the capacitive nature of the electrode configuration, in conjunction with the temporal derivative of the voltage profile. Synek et al. presented a self-assembled current probe with rise time faster than 0.35 ns and a large dynamic range enabling the measurement of electron as well as the much slower ion current components [90]. The utilisation of back current shunts and D-dot and B-dot sensors, respectively, was elucidated in the context of fast pulsed surface DBDs and corona discharges [91–93]. Analysis of statistical distribution of electrical characteristics of microdischarges can be found e.g. in [94–96]. Particular caution is required when inter-

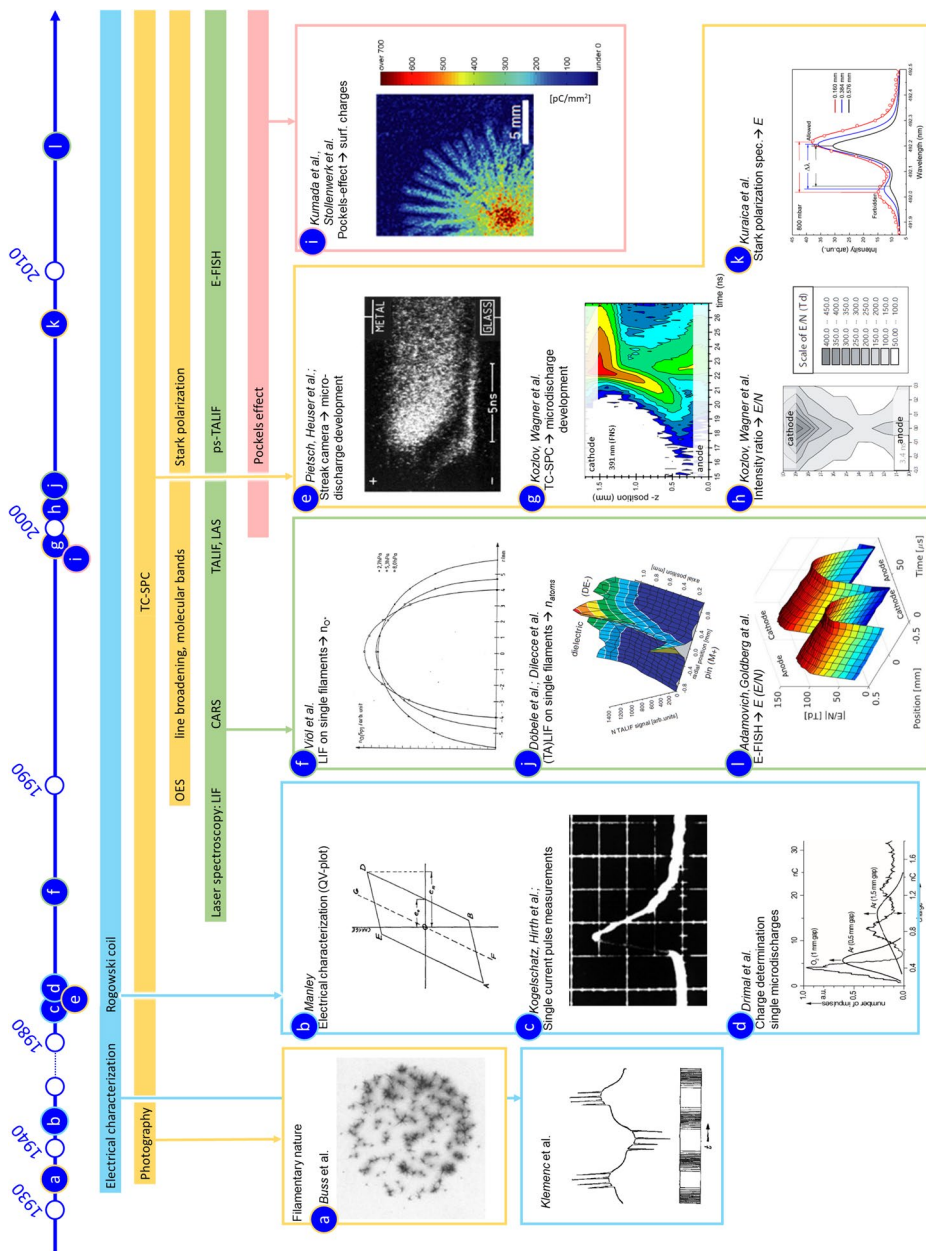


Fig. 3 Selected milestones of barrier discharge diagnostics with figures from [60, 68–79]

preparing current measurements on larger discharge arrangements. Simultaneous individual microdischarges, the occurrence of the above-mentioned oscillations, and undersampling of the signal mean that the properties of these individual discharges cannot be determined

directly. Due to the difficulties involved in measuring current, it is advisable to measure the charge instead and analyse QV-plots for larger DBD arrangements.

Since the 1980s, there has been a focus on conducting fundamental studies on single filaments and the microdischarge properties. These studies require sufficient localisation and stabilisation of the discharge events, which can be achieved using specialised electrode arrangements. Examples include semi-spherical dielectric bodies with embedded electrodes and pin-to-plate electrode configurations, where the pin is embedded in the insulator. The first optical streak camera recordings of such localised microdischarges by Heuser and Pietsch [32, 71, 97] revealed the cathode-directed streamer mechanism in air and air-like gas mixtures. Figure 3 (d) shows the development for a microdischarge in air with 4 mm gas gap (metal anode on top and dielectric covered cathode at the bottom). The fundamental principles of the streamer mechanism were established already in the 1930s [98, 99]. Nevertheless, considerable progress has been made in the last 30 years, as summarised by Nijdam et al. and Cernak et al. [100, 101].

Other notable achievement include the first laser diagnostic investigations of microdischarges by Viöl and Uhlenbusch et al., who used laser-induced fluorescence (LIF) to measure excited oxygen atoms ($^5S^0$) on a localised, pulsed single-filament in oxygen at pressures of 2.7 to 8.0 mbar as early as 1985 [73]. Figure 3 (f) show the radial distribution of the excited oxygen atoms at these conditions. Alongside the electrical and chemical investigations, optical emission spectroscopy (OES) and broadband absorption spectroscopy were applied from the 1990s on, see e.g [102–105]. The deduction of a gas temperature from OES, i.e. rotational band structures of molecules was extensively applied. Bruggeman et al. stated that the equilibrium between translational and rotational degrees of freedom must be ensured prior to any conclusions being drawn regarding the gas temperature. The authors presented a comprehensive analysis of gas temperature determination [106]. Starikovskaia et al. studied fast gas heating in nanosecond pulsed surface DBDs in air and showed that the results of temperature determination in pulsed operated surface DBD can be influenced by the spatial structure of the discharge [50, 91, 107].

Also in the 1990s, further laser spectroscopic work was carried out, e.g. by Ochkin et al. [108, 109] and Uhlenbusch et al. [110] to determine the electric field strength and temperatures using coherent anti-Stokes Raman scattering (CARS). These activities continued until the mid 2010s [111, 112]. Furthermore, Müller and Geroва determined the electron density using Stark broadening in a plate-to-plate DBD in helium. The work of Kuraica et al. applying polarization-dependent Stark splitting and shifting starting in the mid 2000s for the determination of the electric field strength in DBDs with helium in the gas mixture also deserves a special mention [78, 113–116]. Figure 3 (l) shows the splitting of the He I 492.19 nm line recorded at several distances from the cathode in plate-to-plate DBD in helium. The separation between the allowed and the forbidden component is a measure of the electric field strength. As demonstrated by Schulz-von der Gathen and Golda in recent research, the aforementioned method was employed in the context of micro-scaled surface DBD arrays [117–119].

In the mid-1990s, Kozlov et al. introduced the method of time-correlated single photon counting (TC-SPC), also referred to as cross-correlation spectroscopy, for the study of single DBD microdischarges [120]. As the designation suggests, TC-SPC is predicated on the detection of individual photons of a repetitive light signal. For each recorded photon, the detection time is measured and the waveform light signal is reconstructed by accumulation

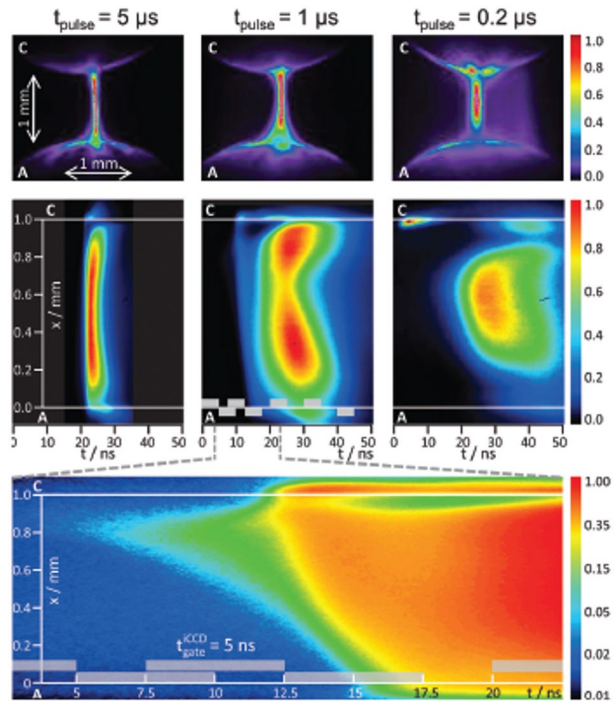
[121]. Consequently, it is the most sensitive technique for the spatio-temporally and spectrally resolved recording of microdischarge development, but also provides a high dynamic range and a high signal-to-noise ratio. The time resolution of TC-SPC is limited by the transit time spread (TTS), rather than by the width of the output pulse of the photomultipliers. The TTS of the majority of detectors utilised is 180 ps, which is considerably higher than the theoretical best resolution stipulated in the majority of publications. Nevertheless, even the TTS-limited time resolution is adequate for the recording of streamer development in discharge gaps of less than 1 mm. Adler and Müller also used SPC to analyse the excimer formation and decay processes on a single-filament in xenon [122]. Sewraj et al. studied temporally resolved emissions in the vacuum ultraviolet, ultraviolet, visible and near infrared range at single-filament DBDs in nitrogen, argon and xenon in correlation with electrical measurements [123–125]. Meanwhile, TC-SPC was successfully applied to coplanar and surface DBDs in air and N_2/O_2 gas mixtures as well as barrier discharges in CO_2 and transient spark discharges [25, 126–137]. The latest realization is the so-called multi-dimensional TC-SPC which enables automatic scanning over the discharge volume [136, 138–140]. The TC-SPC recording of the molecular bands of nitrogen (first negative – shown in Fig. 3 (g) as an example – and second positive system) permitted the spatially and temporally resolved determination of the reduced electric field strength (E/N) in air. This was initially resolved along the microdischarge axis [120], and subsequently in 2D as shown in Fig. 3 (h) [74]. The method has been further developed and is now also applied to other gases [76, 129, 130, 134, 140–144] and pulsed operated surface DBDs [145, 146].

The sensitive and fast imaging and spectroscopic devices such as intensified charge-coupled device (ICCD) cameras and optoelectronic streak cameras have been demonstrated as a reliable method for the documentation of discharge development and/or morphology, and, can now be considered as a kind of standard diagnostics. Simek et al. demonstrated ICCD microscopic imaging on single filaments in coplanar DBD geometry [147–149]. Modern commercially available streak cameras offer time resolutions of up to 2 ps. ICCD and streak cameras even enable the analysis of individual discharge events [25].

TC-SPC as well as ICCD and streak camera measurement have confirmed that the microdischarge development takes place in four phases in sinusoidal as well as most pulsed operated DBDs: (1) The pre- or Townsend phase with the formation of the first electron avalanche generations building up a localized volume charge in front of the anode, (2) the propagation phase with the development of the cathode-directed streamer head (or ionization wave), i.e. a rapid and spatially progressive increase of the local electric field, (3) the glow phase with a continuous and quasi-neutral discharge channel in the volume and the spreading of discharge channels on the insulator surfaces, and, (4) the decay phase initiated by the accumulation of the surface charges. The surface charges remain much longer than the high-voltage period and cause local enhancement of the electric field in the Townsend-phase.

However, charge carriers and/or long lived neutral excited species can also remain in the volume (although on much shorter time scales, namely μs) and affect the discharge formation of the following microdischarge. An example for this was discovered in single-filament DBDs operated with rectangular high-voltage pulses ($dV/dt \approx 250$ V/ns), see Fig. 4. The steep voltage increase results in a shorter Townsend-phase and an increase of transferred charge and dissipated energy [95, 150]. Microdischarges occur at the rising and the falling slope of the high-voltage pulse, thus for short voltage pulse duration (below 20 μs under the conditions being considered) the discharge event at the falling slope is significantly influ-

Fig. 4 ICCD and streak camera recordings of pulsed-operated single-filament DBDs in N_2 with 1000 ppm of O_2 at atmospheric pressure. Shown are the discharge emission structure (top row) and spatio-temporal development (middle and bottom) at the falling slope 5; 1 and 0.2 μs after the discharge event at the rising slope [46]



enced by the remaining volume charges. The Fig. 4 shows synchronized ICCD and streak camera recordings for microdischarges in N_2 with 1000 ppm O_2 admixture at the falling slope of the rectangular high voltage, i.e. after $t_{\text{pulse}} = 5; 1$ and $0.2 \mu s$ after the discharge event at the rising slope [46]. The ICCD images show broader microdischarges and wider surface discharge expansion with decreasing the pulse duration while the streak recordings reveal the spatio-temporal discharge development. In particular, the zoom for the $1 \mu s$ case shows that the starting point of the streamer head shifted towards the cathode by the remaining volume charge, thus, leading to the propagation of a cathode- as well as an anode-directed ionization front. For 200 ns, the breakdown proceeds even without a volume streamer propagation [46, 151]. In this case the 2D emission structure of the microdischarge resembles the structure of the previous discharge event (at rising slope), i.e. it is a re-illumination in its afterglow. These effects of breakdown tailoring are not limited to single-filament DBDs as it was also successfully transferred in multi-filament arrangements recently [152].

The determination of species densities in DBDs at or near atmospheric pressure by LIF and two-photon absorption LIF (TALIF) succeeded in the mid 2000s, e.g. by Döbele and Schulz-von der Gathen et al. [75] and de Benedictis and Dilecce et al. [153–156]. Figure 3 (i) shows the distribution of nitrogen atoms in a single filament as measured by TALIF [75]. Notably, the maximum is obtained in front of the anode, i.e. behind the high-field region of the streamer head in the discharge channel where electron density is higher. The transient character and short duration of microdischarges requires short pulsed lasers for time resolved laser diagnostics such as LIF and TALIF. Picosecond pulsed lasers are used in atmospheric pressure plasma diagnostics since the 2010s and meanwhile there are numer-

ous contributions about laser diagnostics on DBDs on different species such as atoms, radicals or metastable molecules including photo-detachment studies on negative ions, see e.g. [19, 21, 117, 153, 157–168]. Laser Thomson scattering to measure electron densities and temperatures was first applied to nonthermal plasmas in the 1990s (see review paper [169]), but applied to barrier discharges, in particular plasma display panels around 2000 by Muraoka et al. [170–173]. Although widely used for plasma jets [174] its application to barrier discharges is still rare.

Molecular beam mass spectrometry (MBMS) was first applied on DBDs in the 2010s by Benedikt et al. [175] and Bradley et al. [176, 177]. This technique enables the analysis of stable chemical components as well as positive and negative ions. A review about MBMS on atmospheric pressure plasmas, focussing on plasma jets, is given in [178].

Another highlight of DBD research was the first measurements of surface charges on the dielectric barriers by the Pockels-effect, which succeeded first in the mid-1990s by Kumada et al. [77, 179–182] and then with continuing efforts by Komuro et al. [183–189] and Stollenwerk et al. [179, 190–205] where even time resolved measurements with resolution down to 200 ns could be realized. Figure 3 (k) shows the surface charge distribution in a surface DBD arrangement in air during a positive high voltage pulse [77]. This research gave more insights in the understanding of the formation of distinct regular and irregular discharge pattern, see the works of Wertheimer et al. [206, 207] or Stollenwerk and Purwins et al. [201, 202, 208–211]. The research in the field was reviewed recently by Komuro [85]. The drawback of the Pockels-effect is that electro-optic crystals with a higher dielectric constant than usual have to be used as the barrier material. Some crystals have distinct differences from the classical dielectric barrier materials, such as a smaller secondary electron emission coefficient or a photo-induced transport of charges.

A relay high-voltage-circuit-based electrical measurement approach on a surface DBD microplasma array was recently introduced by Golda et al. [212]. Another relatively new approach for the measurement of plasma generated electric field pattern at insulator surfaces is Mueller polarimetry as introduced by Sobota and Guaitella et al. [213–218]. This technique can also monitor temperatures and enables the measurement of polarization characteristics. To date, the application has been restricted to plasma jets; its use on barrier discharges remains to be explored. Reviews on plasma jets and its interaction with surfaces (including dielectric and liquid targets) were given recently by these authors [20, 219].

The electric field induced second harmonic generation (E-FISH) represents arguably the current state of the art for laser based transient electric field measurements [26]. The method was first experimentally investigated in the 1970s and first used for measurements plasmas in 2018 by Goldberg et al. [220]. Most studies are on pin-to-pin plasma or plasma jets, but in the same year the application of E-FISH on surface DBDs started as well. These activities are ongoing [221–223]. As an example, Fig. 3 (l) shows the measured reduced field strength in a sinusoidal operated DBD in nitrogen in the so-called atmospheric pressure Townsend discharge (APTD) regime [76]. This regime is diffuse or uniform due to memory processes such as Penning-ionization, associative ionization or secondary electron emission by metastable species impact [224–228]. Figure 3 (j) shows the spatiotemporal development of E/N over a full high-voltage period. Its value changes with time in agreement with the current profiles, but gradually increases only slightly towards the respective cathode as expected for a Townsend discharge. The E-FISH results were compared with a determination from the molecular bands (first negative and second positive system) intensity ratios,

from an electrical equivalent circuit approach (i.e. via gap voltage determination from externally measured electrical signals) and the Townsend first coefficient (α) from spatial optical emission profiles in [76]. Despite the fact, that the temporal profiles were similar, the best agreement was found between E-FISH and the equivalent circuit approach while intensity ratio and α -coefficient determinations give systematically higher values. Only E-FISH provided a spatially refined electric field in this study. The limitations of applicability and the possible sources of systematic errors of all methods are discussed in detail in this publication. E-FISH is not restricted to certain gases like CARS or emission-based methods like Stark-polarization spectroscopy. Furthermore, it enables the electric field determination also without plasma, e.g. the residual field formed by surface charges or prior to breakdown.

It has to be mentioned that most of the methods were not exclusively developed for barrier discharge studies, but applied first to other nonthermal (atmospheric pressure) plasmas. Up to the 1990s this has been mostly corona and other pin-to-pin/pin-to-plate discharges. But since then, the analysis of plasma jets and fast pulsed discharges lead to a much wider spectrum of diagnostic approaches in the research field. Although some plasma jets can be regarded as similar to DBDs these investigations were not fully included in this overview and only some important links has been set in the text above. However, the main challenge on realizing the diagnostic methods on DBDs and microdischarges is still the adaptation of the plasma source to enable stable and reproducible operation or to reach sufficient sensitivity and/or spatial and temporal resolution.

Summary and Outlook

The preceding section should demonstrate the significant advancements made in the field of barrier discharges and nonthermal plasmas at atmospheric pressure, particularly over the past four decades. A more sophisticated understanding of discharge formation has been achieved, encompassing electric field distribution and development, and the number density and kinetics of selected species, including radicals, metastable, excited, and ionic species (pure and hydrated cations and anions) in DBDs. Recently, there has been an increase in the exploration of the impact of discovered volume and surface memory effects on the breakdown mechanism and phenomenology.

Despite this progress, the diagnostics on these discharges remain as a challenge. In [229] Lu et al. state that “diagnostics are key enablers of advances in both science and technology” and mentions a profound understanding of non-equilibrium energy deposition, the development of novel “single-shot” or real-time diagnostic techniques with high spatial resolution and the establishment of reference sources as the grand challenges in the research field. While the understanding of discharge development in the volume is quite well advanced, the interaction with boundaries and its surface is by far not fully explored. This is definitely a task for the future which will even more take into account more complex barriers such as catalysts or liquids.

Nevertheless, *in situ* diagnostics for volume and surface processes are still required. Although laser absorption spectroscopy has become a popular method for characterizing atmospheric pressure plasmas [230, 231], due to the small scale, i.e. short absorption length, the application is not straightforward. It has been demonstrated that high-finesse optical cavities enhance the absorption path length by at least a factor of thousand. Techniques such

as cavity ringdown spectroscopy have been successfully applied to plasma jets and pin-to-pin discharges [232–234] and DBD reactors [165, 166], but not yet on single-filament DBDs. It can be stated, that the studies and results on single DBDs are representative for the operation of large scale DBD reactors giving important hints on its design, the local energy dissipation as well as the fundamentals of the ongoing physical and chemical processes.

In the near future, it will be impossible to conduct research without the design and implementation of innovative tools and workflows for the structured, automated storage, documentation, processing and analysis of data. Data science methods need to be advanced or developed [235–238], and this will be vital for encouraging data-driven research and development. In particular, the application of machine learning methods and neural networks is expected to have a significant impact, as these are revolutionizing data analysis and creating new, efficient simulation possibilities.

The ongoing “renaissance” of DBDs is due to continuous development of diagnostic capabilities and advancements in simulation efforts. However, it is also driven by societal needs and a wide range of new application needs. The advent of novel nonthermal plasma applications has led to an increasing level of complexity in the field of research. This progression is characterised by the emergence of more complicated electrode configurations, gas flow dynamics, a complex plasma-surface interactions and a wide spectrum of chemical processes. Attention must be given to all these elements to advance knowledge in this field.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval Not applicable.

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